

Taewon Yoo

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RESEARCH INTERESTS

Human-Computer Interaction (HCI), Computer Supported Cooperative Work (CSCW), Social Computing, Visual Analytics, Human-Centered AI, Sustainable AI

EDUCATION

Kyung Hee University

Bachelor of Software Convergence (Data Science)

Yongin, Republic of Korea

Mar. 2017 - Aug. 2024

WORK EXPERIENCE

Research Intern, University of Seoul

Human-Centered Artificial Intelligence Lab (HCAIL)

Seoul, Republic of Korea

Jun. 2020 - Jun. 2024

- Developed a visual analytics tool supporting ML developers to consider carbon emissions in development
- Developed a smartphone application for dietary assessment for older adults
- Advisor: Prof. Hyunggu Jung

Data Scientist, AUTOCRYPT (alternative military service)

Data Processing Team

Seoul, Republic of Korea

Aug. 2021 - Jul. 2023

- Conducted data analysis of mobility app services to enhance usability and increase user engagement
- Conducted data analysis of EV, delving into battery status, sensor performance, and motor functionality
- Developed a real-time application to calculate taxi fares using geospatial data

Research Intern, KIST Europe Forschungsgesellschaft mbH

Smart Convergence Group

Saarbrücken, Germany

Aug. 2019 - Jan. 2020

- Developed and evaluated a generative adversarial network to solve data-imbalance problem
- Advisor: Dr. Sungho Suh

PUBLICATIONS

International Journal Papers

1. S. Na, S. Park, J. Jun, H. Jeong, **T. Yoo**, M. Chun, H. Lee, H. Lee, H. Jung. “SUV: Students’ Understanding Visualizer to Support Instructors in Synchronous Online Lectures”. *IEEE Access*, August 2023. DOI: 10.1109/ACCESS.2023.3303030
2. M. Chun, H. Jeong, H. Lee, **T. Yoo**, H. Jung. “Development of Korean Food Image Classification Model Using Public Food Image Dataset and Deep Learning Methods”. *IEEE Access*, December 2022. DOI: 10.1109/ACCESS.2022.3227796

International Conference Papers

1. **T. Yoo**, H. Lee, S. Oh, H. Kwon, H. Jung. “Visualizing the Carbon Intensity of Machine Learning Inference for Image Analysis on TensorFlow Hub”. In *Companion Publication of the 2023 Conference on Computer-Supported Cooperative Work and Social Computing (CSCW’23 Companion)*. ACM, Minneapolis, MN, USA, October 2023. DOI: 10.1145/3584931.3606959
2. **T. Yoo**, M. Chun, Y. Bae, S. Kwon, H. Jung. “Exploring the Community of Model Publishers on TensorFlow Hub”. In *Companion Publication of the 2022 Conference on Computer Supported Cooperative Work and Social Computing (CSCW’22 Companion)*. ACM, Virtual, October 2022. DOI: 10.1145/3500868.3559477
3. **T. Yoo**, H. Jeong, D. Lee, H. Jung. “LectYS: A System for Summarizing Lecture Videos on YouTube”. In *Proceedings of the 26th International Conference on Intelligent User Interfaces (IUI’21 Companion)*. ACM, Virtual, April 2021. DOI: 10.1145/3397482.3450722
4. **T. Yoo**, H. Jung. “Category Classification of Educational Videos on YouTube through Machine Learning Approaches using Video Titles”. In *Proceedings of the US-Korea Conference on Science, Technology and Entrepreneurship (UKC)*, Chicago, IL, USA, August 2019.

Domestic Conference Papers

1. **T. Yoo**, H. Jung. “Exploring the Community of Sustainable AI Developers on Hugging Face”. In *Proceedings of the HCI Korea 2024 Conference*, Hongcheon, Republic of Korea, January 2024. **(BEST PAPER AWARDS)**

2. **T. Yoo**, D.Lee. “Video Summarization with Unpaired Dataset and Adversarial Network”. *In Proceedings of Korea Software Congress (KSC) 2023*, Busan, Republic of Korea, December 2023.
3. **T. Yoo**, H. Jung. “Category Classification of Educational Videos on YouTube through Machine Learning Approaches”. *In Proceedings of Korea Computer Congress (KCC) 2019*, Jeju, Republic of Korea, June 2019.
4. **T. Yoo**, H. Jung. “Needs Analysis Based on Vlogging Career”. *In Proceedings of the HCI Korea 2019 Conference*, Jeju, Republic of Korea, February 2019.

ACADEMIC SERVICES

Reviewer

CSCW'24 Poster

Student Volunteer

CHI'24, CSCW'21 - '23, RecSys'21, '22, FAccT'22, IUI'21, ASSETS'21, DIS'21

PRESENTATIONS

Sustaining Scalable Sustainability Workshop, CHI'24

Visualizing the Carbon Intensity of Machine Learning Inference for Image Analysis on TensorFlow Hub

EXTRACURRICULAR ACTIVITIES

Teaching Assistant

SWCON104 Web and Python Programming, Kyung Hee University

Spring 2019, 2020, 2021

Course Assistant

GEE1952 Major Exploring Program, Kyung Hee University

Fall 2018

GEE1920 Freshmen Seminar, Kyung Hee University

Spring 2018, 2019

Vice President of Student Association

Department of Software Convergence, Kyung Hee University

2018

HONORS AND AWARDS

Best Paper Awards

100,000 KRW, HCI Korea 2024

2024

Excellence Prize

5,000,000 KRW, Hyundai AutoEver (Barrier Free App Development Contest)

2022

The National Scholarship for Science and Engineering

Full tuition fee for 4 semesters (17,160,000 KRW), Korea Student Aid Foundation

2019

The National Scholarship for Excellence in Software

12,000,000 KRW, National Program of Excellence in Software

2019

Encouragement Awards

Blue-print Ideathon

2019

Humanitas Scholarship

2,100,000 KRW, Kyung Hee University

2018, 2019

Bright Society Scholarship

4,000,000 KRW, Kyung Hee University

2018

SKILLS

Programming: Python, Kotlin, Dart, Java, and Javascript

Data Science: Data Visualization & Insights, Customer Data Analysis and Predictive Modeling

Applications: SQL, Docker, Kubernetes, L^AT_EX, Flutter, Spring Boot, and Miro